6SENG010C · SOFTWARE ENGINEERING FINAL PROJECT · PROJECT PROPOSAL

Market-State-Conditioned Orchestration of Heterogeneous Agents for Financial Decision Support

How should the influence of methodologically unlike financial agents be set when the reliability of their evidence channels varies with market conditions? I design, implement and evaluate a market-state-conditioned orchestration operator on Colombo Stock Exchange data.

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PROPOSAL STORYLINE

A research spine and a software engineering spine, carried together

Research spine	Software engineering spine	Where it is
Agent reliability varies with market state; reviewed systems combine as though it does not	A pipeline that does not emit confident output from an empty channel	Slides 3, 9
59 reviewed papers, the six nearest systems contrasted clause by clause	Every requirement carries the paper or the measurement that forced it	Slides 4, 5, 6, 8, 10
One main question, two supporting, each answerable by a planned experiment	Four agents, one orchestrator, nine stated exclusions	Slides 6, 7, 10
Design science, controlled comparison, five arms, one ablation per state input	Five layers, point-in-time access, a verified data pipeline	Slides 11 to 15
An operator with a named lineage, plus evidence on whether it earns its place	Research prototype, no interface, three decisions changed by responsible practice	Slides 16 to 20

Spoken: I will show the gap and the contribution, then how the artefact, methodology and evaluation address it, then what I am not claiming.

PROBLEM AND GOAL · CRITERION 1, 10 PER CENT

Influence over a shared decision is fixed at design time

Four agents produce signals, and something must turn them into one decision. In the systems I reviewed that combination is set once and never moves, and in most of them there is no weight object at all to move.

Evidence of significance

Three documented failures: a CSE study fixing a 3:1 split on an easier task's score; an optimiser fed uncalibrated model-authored returns; a pool degrading 10.36 points through a fixed prompt.

Who is affected

The investment decision-maker acting on the output, the researcher running the comparison, and retail participants least able to absorb a cost a model assumed away.

Project goal

Build the operator and produce attributable evidence on whether conditioning earns its place against fixed, static learned and performance-reactive weighting. Not on whether the system is profitable.

Expected outcomes

A pipeline where changing liquidity changes the weights, a five-arm comparison with ablation deltas, a CSE baseline set, and a stated position on confidence comparability.

Four lines of evidence, one gap they converge on

Evidence source	What it shows	Limitation or tension	Resulting gap
Academic literature, 59 papers	Heterogeneity is widely adopted; combination is treated as a solved detail	Several systems carry no weight object. Those that do weight assets, not influence	Influence is fixed at design time and adapts to nothing measured outside the agent
Existing systems	Four named systems each get one necessary clause right	The nearest neighbour in each direction is strong, so the contrast must be clause by clause	I have not found a system that combines all four clauses at once
Domain evidence, the CSE	Volume Granger-causes the index at p = 0.0041; macro variables reach nothing	Channels go empty for days, so equal trust across agents is false here	A setting where channel reliability genuinely varies, and can be measured
Forecasting literature	Feature-conditioned combination learns weights over a heterogeneous pool	Its features are properties of the series, computed once from history	A generalisation along one identified axis, stated as such

Fifty years of evidence says equal weighting is hard to beat, which is why it is the pre-registered primary benchmark rather than a convenience baseline.

Three themes, and what each one forces on the design

THEME 1 · THE MARKET

Thin trading is the condition here, not an edge case

Volatility clustering in three CSE studies, long memory across all three regimes, and illiquidity shocks depressing returns.

The conditioning variables are locally evidenced, not borrowed

THEME 2 · COMBINATION

Every necessary clause exists somewhere, in a different system

Analyst pipelines, feature-conditioned combination, continuous state conditioning, and explicit weight vectors over assets.

The gap is the conjunction, so the contribution is a generalisation

THEME 3 · EVALUATION

A strong minority evaluates well, and almost none can be rerun

Walk-forward, regime decomposition and cost sweeps appear. Of my 59 papers, 4 are reproducible from the paper alone.

Cached outputs across arms, a pre-registered margin, a released pipeline

Counts are of my own 59-paper corpus, not of the field. The classification rule that produced them is in the backup.

Every alternative also produces weights. The question is what the weights may depend on

WHAT EXISTS

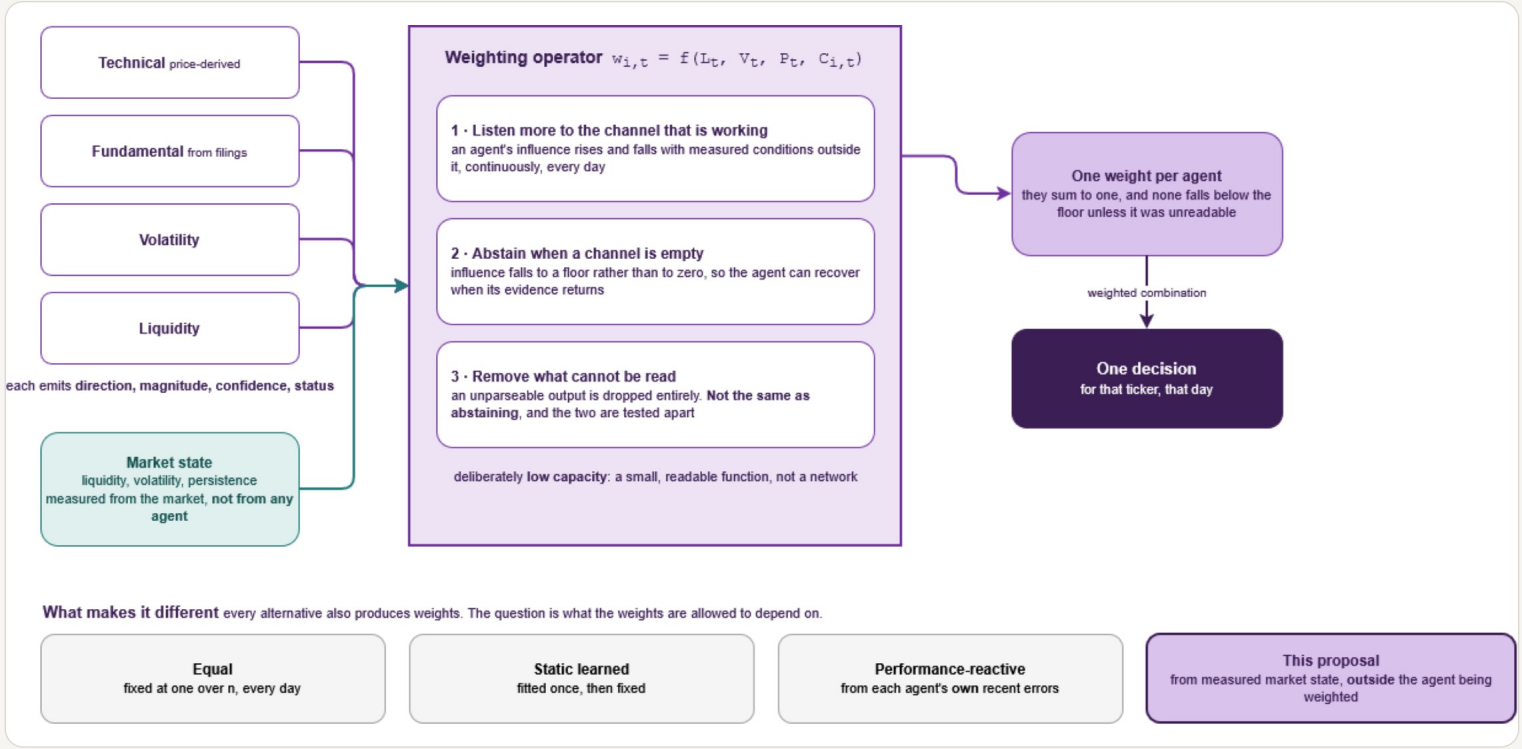
Multi-agent systems that combine analysts, mostly through a prompt. Influence fixed at design time.

WHAT IS MISSING

A weight that depends on conditions measured outside the agent being weighted, recomputed every step.

THE CONTRIBUTION

Attributable evidence on whether state-aware orchestration earns its place, positioned as a generalisation of feature-conditioned combination along one named axis.



Not a duplicated implementation. Three claims were withdrawn once I found papers that already did them, and the withdrawals are on the record.

No question is answerable yes or no. Each asks how

MAIN RESEARCH QUESTION

How should the influence of heterogeneous agents be adapted to measured market state, and under what conditions does adaptation earn its place?

Answered by five arms over byte-identical cached outputs, with one ablation per state input. Success: the median per-fold gain exceeds the interquartile range of the equal-weight arm across 16 folds, and clears HAC and bootstrap tests at 0.05.

SUPPORTING QUESTION 1

How should the system alter execution and the influence of price-derived agents when liquidity deteriorates?

Sensitivity sweep of weights against liquidity, with two ablations that separate trust from execution.

SUPPORTING QUESTION 2

How can confidence from unlike architectures be placed on a common scale?

Four structural mechanisms, each mapped through its own accuracy-at-threshold curve. If they do not land in one unit, that is the finding.

OBJECTIVES

Build the artefact, then evaluate it honestly.

Data layer, four agents behind one contract, state vector, operator, five-arm harness. Acceptance criteria A1 to A6. A null at the pre-registered margin is acceptable, decided in advance.

Each question has a pre-specified response if the answer is unfavourable, written before any run.

The nearest system in each direction gets one clause right

System or baseline	What it gets right	Where it stops	What that forces on my design
Prompt-combined agent systems	Heterogeneous roles feeding one decision	They combine through free text, so no weight object exists	A common structured output contract has to come first. FR2
HedgeAgents	Optimisation over an explicit weight vector	The weights are over assets, so which method to trust cannot arise	The single-shared-decision clause, and calibration before any optimiser
MacroHFT	Continuous conditioning on market state, weight trace published	Six identical networks, so conditioning without heterogeneity	The heterogeneity clause, with entropy collapse as an observed failure mode
Feature-conditioned combination	Learned weights over a heterogeneous pool, closest in form	Features are properties of the series, computed once from history	Named as the lineage of this work rather than a rival
Pavithya et al. (2021), on the CSE	Two unlike models, one decision, on this market	A fixed 3:1 split taken from a score on an easier task	The motivating failure: a weight cannot come from an out-of-domain score

The proposal is compared against four alternatives over identical cached outputs, plus an oracle fitted on the test window held as a hindsight ceiling, not an achievable arm.

STAKEHOLDERS, USERS AND CONTEXT OF USE · CRITERION 3, 20 PER CENT

The exposure is not confidentiality. It is false confidence

PRIMARY OPERATOR · THE RESEARCHER

Needs reproducible runs and comparisons that are actually fair.

Evidence: a published comparison gave its baseline less scaffolding than the proposed system. Design: byte-identical cached outputs across all five arms, enforced architecturally.

MOTIVATING BENEFICIARY · THE INVESTMENT DECISION-MAKER

Needs a decision that does not carry confident influence from an empty channel.

Evidence: five documented cases of reliability varying unnoticed, including a 10.36-point loss. Design: the abstention channel and the confidence gate.

THE EXCHANGE AND ITS DATA REGIME

Licensed data must be used within its terms.

Evidence: the purchased set carries 46 securities absent from the API, a 12.8 per cent gap. Design: official licensed extraction as the base panel, licence recorded per row.

WIDER EXPOSURE · RETAIL PARTICIPANTS

Should not be exposed to a system that assumes away the cost they would pay.

Evidence: institutional ownership at 66.1 per cent against 5.7 per cent insider. Design: cost inside the allocation objective, and the artefact framed as a prototype rather than a recommender.

A well-formed directional signal produced from a channel that is empty today is the failure this system is designed to make visible.

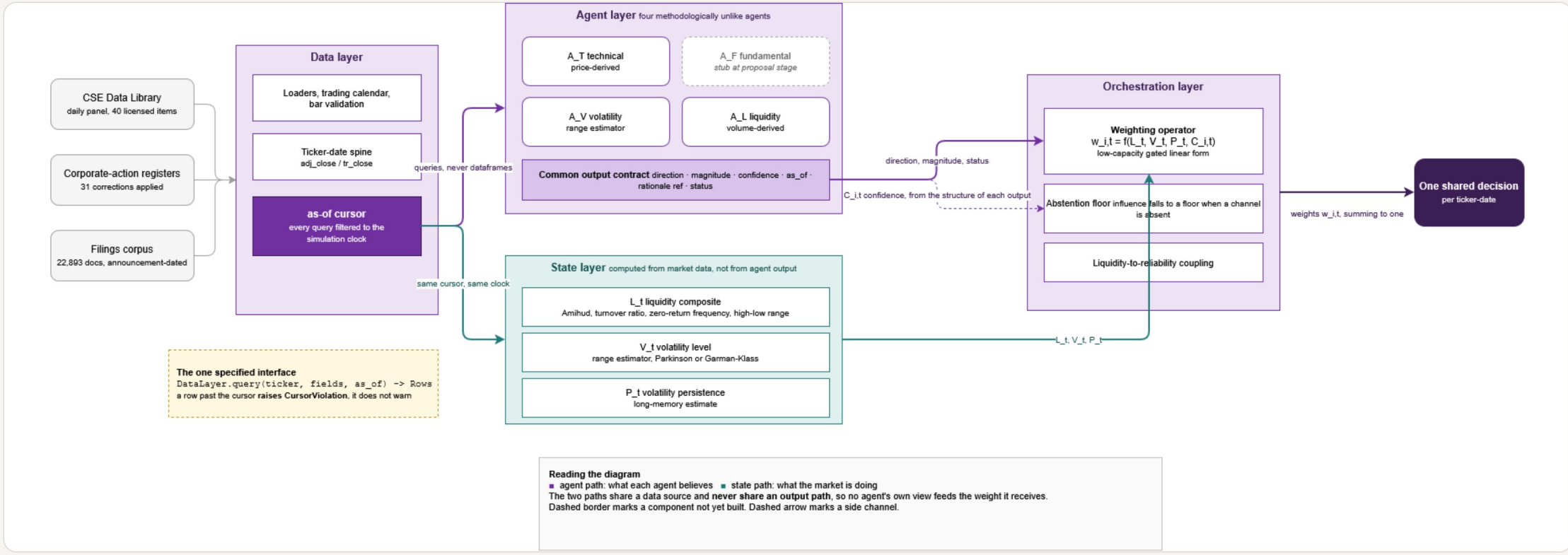
REQUIREMENTS ENGINEERING · CRITERION 3, 20 PER CENT

Every requirement carries the evidence that forced it

ID	Requirement	Type	Source or rationale	Acceptance evidence
FR1	The data layer shall refuse any row past the simulation clock, raising rather than warning	Functional	Two documented point-in-time failures on this market	Cursor-violation test and an exception log
FR2	Every agent shall emit the common output contract, all six fields	Functional	Prompt-combined systems in which no weight object exists	Contract conformance across all four agents
FR10a	The system shall emit the weight entropy series and the liquidity sensitivity	Functional	Acceptance criteria A2 and A3 are defined on these two quantities	Entropy series and sensitivity sweep plot
NFR7	Outputs, state and weights shall be logged one row per ticker-date	Quality	An explanation that was not recorded is not an explanation	A sampled decision rebuilt end to end from the log
DR10	Ex-date reliability shall be carried as a field, with inferred dates flagged	Data	No documented date in the registers is an ex-date; 259 are approximate	ex_date_basis populated; affected days flagged
B2	No fifth agent, including a macroeconomic agent	Boundary	Rejected on evidence: adjusted R-squared 0.189 across five macro variables	Scope statement carrying the measurement

43 requirements: 11 functional, 7 non-functional, 10 data, 7 model and AI, 8 boundary. Exactly three are Should. Six shown here; the full traceability matrix is backup slide B3.

Five layers, one clock, and no agent feeding the weight it receives



One interface is fully specified: DataLayer.query(ticker, fields, as_of) returns rows and raises CursorViolation. Python and the scientific stack; the model backbone stays open against four stated selection criteria.

Purchased, reconciled, and honest about the one gap

Data need	Source and route	Quality check, and its status	Ethics and licence control
Price and volume panel	Official CSE Data Library, purchased. 355 core securities, 233 candidates, July 2012 to June 2026	Verified: turnover reconciled at 0.13 per cent, 99.93 per cent close agreement against an independent archive	Licensed purchase, not scraping. Licence recorded per dataset row
Corporate actions	Data Library registers. Adjusted panel constructed 23 October, carrying adj_close and tr_close	Verified: applied per security per year, confirmed against the next year end. 31 corrections	Official registers rather than inferred adjustments
Liquidity inputs	Turnover, trade count, quarterly float and crossings, held across the full span	Verified: crossings run 8.8 to 47.4 per cent of annual turnover; stripped and unstripped both reported	Same licensed source
Filings corpus	22,893 documents, January 2012 to October 2026, roughly 340 counters	Downloaded and probed, not yet validated for answerability. Every filename carries its announcement date	Exchange terms respected; retained for research, never redistributed
Not held: bid, ask, depth	Absent from all forty purchased items. An exchange request is the only route	The cost work runs on proxies, declared in advance as the plan rather than as a fallback	The released pipeline is code plus an acquisition route, never the data

The window is anchored to an index-regime boundary rather than to data availability: the S&P SL20 launched in June 2012, so the large-cap methodology is constant across the sample.

RESEARCH AND DEVELOPMENT METHODOLOGY · CRITERION 4, 15 PER CENT

Two methodologies, answering two different questions

Research design How the knowledge claim is investigated	Design science research. The artefact is the instrument of inquiry, not the deliverable.
Development process How the artefact is built iteratively	Evolutionary prototyping managed through stage gates. The prototype validates mechanism feasibility, not requirements.
Evaluation protocol How evidence is collected and interpreted	Rolling walk-forward: 1,250-day train, 125-day test, 16 folds, periodic refit. Five arms, regimes decomposed, costs swept.
Validity controls How bias and weak inference are reduced	Point-in-time enforcement that raises rather than warns, identical outputs across arms, margin and test fixed before the run, and a null pre-registered as an acceptable result.

Rolling windows rather than expanding ones, because the decision margin is scaled to fold-to-fold dispersion and an expanding window would shrink it artificially.

Only the weighting function differs between arms

Element	Design
Comparators	Five arms over byte-identical cached outputs: equal weight as the pre-registered primary, static learned, inverse-MSE with no state, the proposal, and an oracle fitted on the test window as a hindsight ceiling
Metrics	Cost-adjusted return over the full window, abstained days held flat, with per-traded-day as a secondary diagnostic. Every directional row carries class balance and Matthews correlation
Protocol	Rolling walk-forward, 16 folds, five seeds with dispersion reported, decomposed across the Easter attacks, COVID and the 2022 crisis, with a cost sweep
Ablation	One ablation per state input, which is what makes the answer attributable. Plus the fundamental agent on and off, two separating ablations, and a stability sweep
Success criterion	The median per-fold gain over equal weighting must exceed that arm’s own interquartile range across the 16 folds, and clear HAC and bootstrap tests at 0.05. A null is acceptable

What this cannot support: accuracy alone says nothing about decisions; no live-trading claim, because slippage is simulated and unvalidated; nothing generalises beyond one market, window and universe.

What runs today, and what is scheduled and says so

Area	Purpose	Evidence, and its status today
Unit and component	Verify isolated logic, estimators and the calibration transform	Planned, none exist today. Targets named and scheduled against the project stages
Integration	Verify end-to-end flow, and that every arm consumes the same cache	Planned. Acceptance criteria A1 and A5, through byte-identical output comparison across five arms
Requirements-based	Check each Must requirement is satisfied	Specified, not yet run. Nine named tests: cursor violation, contract conformance, state routing, simplex invariant, arm isolation, seed determinism, calibration, gating, result schema
Data and model checks	Detect leakage, bad data, drift and degenerate behaviour	Executed. Turnover reconciled at 0.13 per cent, 99.93 per cent archive agreement, 99.82 per cent of VWAPs in range, a validated 3,920-day calendar. These are the regression baselines
Behavioural diagnostics	Detect failures that make a correct-looking result meaningless	Designed. Weight entropy, liquidity sensitivity, agent error correlation, collinearity, weight trajectory, oracle gap, quiet against stressed
Usability, accessibility, security	Assess responsible use and quality constraints	Mostly not applicable, and said so. No interface, no accounts, no personal data. The real exposure is false confidence, tested functionally above

Coverage percentage is not reported: it counts lines executed, while the risk here is a correct-looking result produced from a wrong assumption.

Where a tool helped, and how each use was checked

ITERATION AND FEEDBACK

Weekly panel sessions and fortnightly supervisor meetings, recorded and transcribed. Every point splits into four things that cannot merge: the quote, my reading labelled as mine, the action, the status. Silence is recorded as unanswered, never as approval.

CHANGE CONTROL

Four mechanisms in use: a dated freeze, dated revision headers, corrections marked in place, and superseded text retained beneath rather than deleted. Four worked examples, including one where the correction weakened my own argument.

Activity	Division of labour, and how it was verified	Evidence retained
Reviewing 59 papers	Every paper read, every note written against the source. What each paper claims and where it is weak is my reading, labelled as mine. The paper wins any conflict	59 notes, the validation prompt, claim and evidence and reading kept separate
Building the matrices	Moving 1,621 quoted spans is transcription at scale, which is where a tool earns its place and exactly why it cannot be trusted unchecked. The audit is the point	101 spans unquoted, 174 marks moved, all 1,621 traced back to a note
Deriving the weighting form	Term by term mapping to the papers supporting each term, with my own reasoning flagged as mine	Four terms mapped. The confidence term is named as the thinnest
Orchestration code	Stated intention, not practice. That code does not exist yet, so nothing about it has been verified	None yet, and this slide says so rather than implying otherwise

The data pipeline is verified by external reconciliation rather than inspection: against published annual data and against an independent archive.

PROJECT PLAN, MILESTONES AND DEPENDENCIES · CRITERION 5, 10 PER CENT

Five stages, each gated by the thing it cannot start without

Now to 5 Nov 2026	PP to IPD, 28 Jan 2027	IPD to feature complete	Testing and evaluation	Report, to 1 Apr 2027
Close the SE spine	**Data layer and agents**	**Full build**	**Five arms at full scale**	**Analysis and thesis**
Reconcile data counts, build and rehearse the presentation.	Liquidity profiling, the four agents behind one contract, the orchestrator.	Integration, the cost term inside the objective, the persistence experiment.	Ablations, regimes, significance tests and stability sweeps.	Limitations, write-up and the demonstration.
Submitted slides and the signed Ethics Application	Working prototype. Every Must requirement has a test that runs	Feature-complete artefact, no Must requirement untested	Evaluation results and the CSE baseline set	Thesis, artefact and code, reproducible from a clean environment

Fifteen per cent of each stage is held unallocated against failed experiments and data correction: 43 of 288 hours to the interim demonstration, 32 of 216 after it. Droppability is declared in advance, the persistence experiment first, the orchestration build never.

The hard dependency: the as-of cursor is built before any agent, because it cannot be retrofitted. Implementation is set aside by supervisor instruction until after submission.

The two largest risks threaten the result, not the build

Risk	Likelihood, impact, owner	Mitigation	Contingency
Equal weighting wins	High, high. P1 critical. Mine	Equal weight is the pre-registered benchmark, with the margin fixed before the run	A null is acceptable, decided in advance and reportable
State conditioning adds nothing beyond reacting to recent agent performance	Moderate, high. P2. Mine	The inverse-MSE arm exists specifically to detect this	If that arm matches, performance reactivity suffices here, which is a finding
No documented date in the registers is an ex-date	Certain, moderate. P2. Mine	Reliability carried as a field, 259 events marked approximate	Daily-return models drop or flag those days, and the count is reported
Bid, ask and depth never arrive	High, moderate. P2. Owner: the exchange	Two data levels declared in advance, before either request was made	The proxy route is the plan, and is guaranteed feasible
Available hours overestimated	Moderate, high. P2. Mine	Hour budgets per increment, with fifteen per cent held as reserve	A declared drop order. The build has already been re-cut once
Lookahead enters the pipeline	Low, high. P2. Mine	The as-of cursor raises rather than warns, and is built before any agent	Not retrofittable, which is why it is protected absolutely

Thirty-four scored rows across research, data, technical, schedule and evaluation risk, each with likelihood, impact, priority and owner. Six shown. Three risks are not mine to control, which is the point of the owner column.

Three decisions that responsible practice actually changed

Area	The issue, specific to this project	Decision taken
Ethics and participants	No human participants, no personal data, no primary collection	Removing sentiment removed the whole exposure: platform terms, annotation and redistribution went with it
Legal, IP, data protection	Licensed exchange data, and benchmark sets with differing licences	Purchased rather than scraped, licence recorded per dataset row. The released pipeline is code and an acquisition route, never the data
Security and safety	Not confidentiality. False confidence: a well-formed signal from an empty channel	A prototype, not advice. Abstention and calibration reduce this and do not remove it, because both fail towards overconfidence
Accessibility and EDI	No interface, so no interface surface. What applies is representational	The universe is stratified rather than selected from the top, and the count the screen removes is reported as a statistic
Sustainability and professional practice	Compute cost, and the obligation not to overclaim	Low-capacity form, no reinforcement learning, one agent pass serving five arms. Withdrawn claims recorded, nulls reported

Sentiment removed, which removed the participant exposure. A measurement rejected because its output could not be checked, costing half a sub-gap. Data purchased rather than scraped.

SCOPE, LIMITATIONS AND CLOSING DEFENCE · CRITERIA 7 AND 9

Substantial, bounded, and clear about what it cannot claim

SUBSTANTIAL SE WORK

A point-in-time data layer over a market with no clean panel: five registers joined through a name crosswalk, ten source items changing meaning midway. An evaluation harness, not a backtest.

FEASIBLE BOUNDARIES

Nine exclusions, each with a reason: sentiment, any fifth agent, microstructure execution, a spread estimator, a network orchestrator, a user interface, and profitability as an acceptance test.

WHAT SURVIVES EITHER RESULT

Whether confidence can be made comparable across unlike architectures, a measured cost-curve shape, and a worked case of how to read combination weights.

CLOSING DEFENCE

This project addresses how the influence of heterogeneous agents should be set over one shared decision when the reliability of their evidence channels varies with market conditions.

It is needed because in the systems I reviewed that influence is fixed at design time, and in most there is no weight object at all, while five documented cases show reliability varying unnoticed.

The scope is credible because boundaries are stated with reasons, the hardest benchmark is pre-registered, a null is acceptable in advance, and both live data risks had fallbacks declared before either one bit.

Sources behind the claims on these slides

Ardia, D., Guidotti, E. and Kroencke, T.A. (2024) 'Efficient estimation of bid-ask spreads from open, high, low, and close prices', *Journal of Financial Economics*, 161, 103916.

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Dissnayake, D.M.T.R. and Nanayakkara, N.S. (2025) 'Role of market liquidity in sentiment-based return predictions: evidence from Sri Lanka', *International Journal of Economics and Financial Issues*, 15(2), pp. 201-210.

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Gibbs, I. and Candès, E. (2021) *Adaptive conformal inference under distribution shift*. arXiv preprint.

Li, X., Zeng, Y., Xing, X., Xu, J. and Xu, X. (2025) 'HedgeAgents: a balanced-aware multi-agent financial trading system', *Companion Proceedings of the ACM Web Conference 2025*, pp. 296-305.

Mohamed Riyath, M.I. (2026) 'Modeling long-term volatility memory dynamics in the Colombo Stock Exchange', *IIM Ranchi Journal of Management Studies*, 5(1), pp. 21-45.

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Samarawickrama, I.D.W. and Pallegedara, A. (2023) 'Sri Lankan stock market volatility analysis: an ARMA-GARCH approach', *Sri Lankan Journal of Business Economics*, 12(1).

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Xiao, Y., Sun, E., Luo, D. and Wang, W. (2025) *TradingAgents: multi-agents LLM financial trading framework*. arXiv preprint arXiv:2412.20138.

Zong, C., Wang, C., Qin, M., Feng, L., Wang, X. and An, B. (2024) 'MacroHFT: memory augmented context-aware reinforcement learning on high frequency trading', *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 4712-4721.

Twelve of 59 reviewed sources, selected as the ones carrying claims on these slides. The full list, the CSE Data Library items cited item by item, and the pinned library versions are in the submitted reference list.

BACKUP AND QUESTION SUPPORT

Evidence held ready, and what each piece defends

B1 · The weighting operator, term by term
Four terms, each mapped to the paper supporting it, with my own reasoning flagged. A panel member has already asked this.

B2 · End-to-end pipeline and verification status
Defends feasibility, and separates what is done from what is designed and not yet built.

B3 · Requirements traceability
The chain from source to requirement to test to evidence, across all 43 requirements.

B4 · Arms, margin and the decision rule
Defends the evaluation design, the oracle as a ceiling, and the interquartile decision rule in full.

Also held: the literature and gap matrix
The clause-by-clause contrast against six named systems, and the coding rule behind every count quoted.

Also held: dataset construction and reconciliation
The reconciliation figures are the persuasive part. Plus the sentiment exclusion defence and the full reference list.

Three questions have no good answer yet, and they are named as such rather than deflected.

BACKUP B1 · THE WEIGHTING OPERATOR, TERM BY TERM

Four conditioning terms, each attributed, one flagged as mine

$$w_{i,t} = f(L_t, V_t, P_t, C_{i,t})$$

A low-capacity gated linear form, not a network. Weights are non-negative and sum to one, and none falls below the abstention floor unless its channel was unreadable. The conditioning set is provisional, not settled.

L_t

Liquidity composite

Amihud illiquidity, turnover ratio, zero-return frequency and the high-low range, computed from market data. Recipe attributed to the microstructure literature.

V_t

Volatility level

A range estimator, Parkinson or Garman-Klass. Chosen over a GARCH fit because it needs no in-sample estimation at each step.

P_t

Volatility persistence

A long-memory estimate. Included because long memory is measured locally on this exchange across all three regimes, not assumed from elsewhere.

$C_{i,t}$

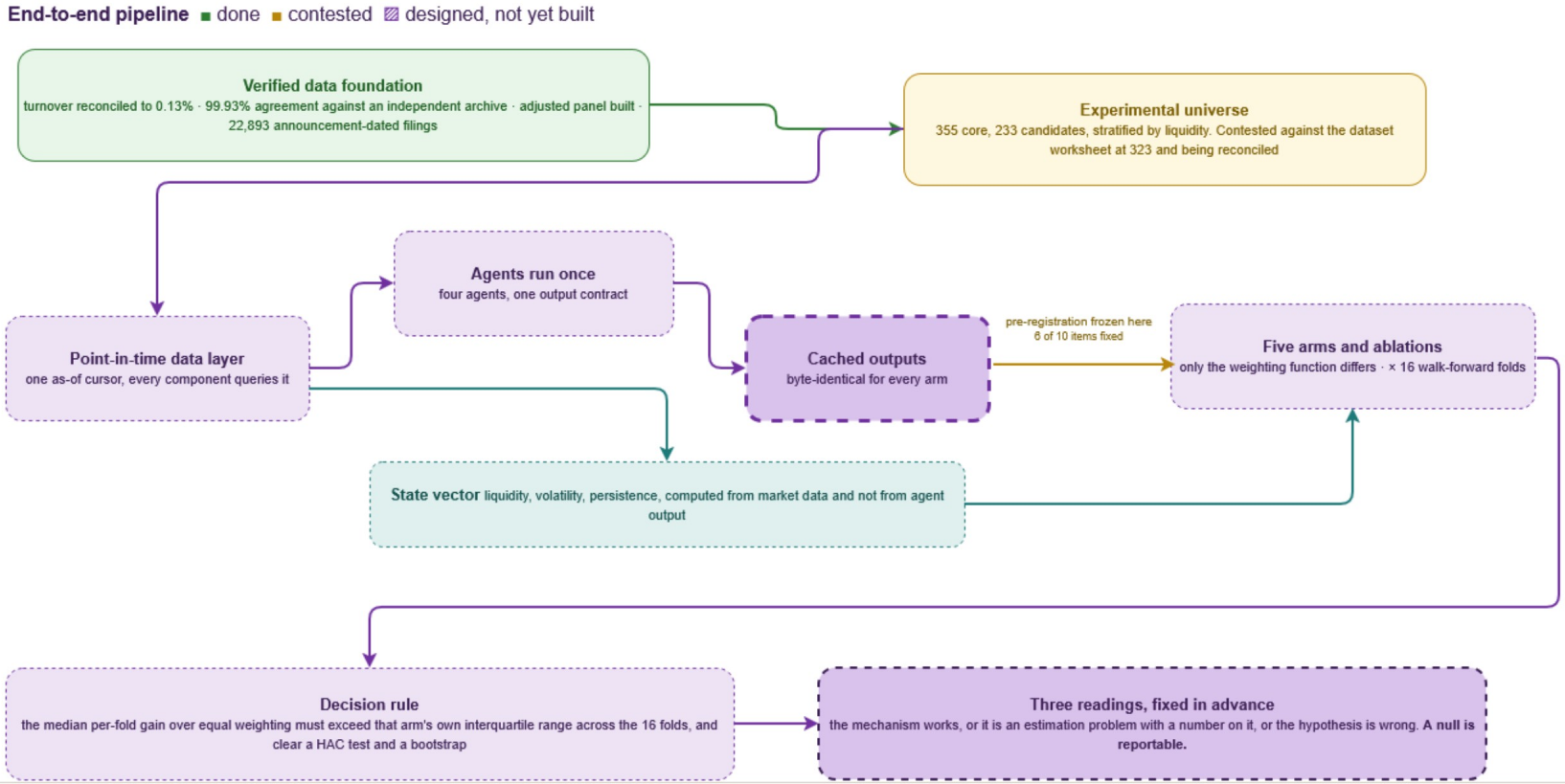
Agent confidence

The thinnest term, and named as mine. Four unlike architectures produce confidence differently. Comparability is the subject of SQ2, mapped through accuracy-at-threshold curves, with conformal calibration as the fallback.

The state vector is computed from market data and never from agent output, so no agent's own view can feed the weight it receives. That separation is architectural, not a convention.

BACKUP B2 · END-TO-END PIPELINE AND VERIFICATION STATUS

Done, contested, and designed but not yet built



BACKUP B3 · REQUIREMENTS TRACEABILITY

43 requirements, and the chain each one sits in

11	7	10	7	8	3
Functional	Non-functional	Data	Model and AI	Boundary	Should, and only three

THE TRACE CHAIN, HELD FOR EVERY ROW

Source evidence, the paper or the measurement that forced it, then the requirement, then the acceptance evidence, then the test that produces that evidence, then the research question it serves.

A requirement with no source is not in the set. A requirement with no acceptance evidence is not a Must.

THE NINE NAMED REQUIREMENTS-BASED TESTS

Cursor violation · contract conformance · state routing · simplex invariant · arm isolation · seed determinism · calibration · gating · result schema

Specified, not yet run. Scheduled against the stages on slide 17.

Three Shoulds only: FR9, FR10b and MAI7. Everything else is a Must or a stated Boundary, which is what keeps the scope defensible under pressure.

BACKUP B4 · ARMS, MARGIN AND THE DECISION RULE

Five arms, and what each one is there to rule out

Arm	What the weights depend on	What it rules out, or establishes
Equal weight · pre-registered primary	Nothing. One over n, every day	The hardest benchmark in the forecasting literature. If this wins, that is the reported result
Static learned	Fitted once on the training fold, then fixed	Whether any differentiation helps at all, before adaptation is considered
Inverse-MSE	Each agent’s own recent errors. No market state	Whether exogenous state adds anything beyond reacting to recent performance. The sharpest rival
The proposal	Liquidity, volatility, persistence and confidence, measured outside the agent	The claim under test
Oracle · hindsight ceiling	Fitted on the test window itself	Not achievable, and not a comparator. It sizes how much headroom any weighting scheme could have had

THE DECISION RULE, FIXED BEFORE ANY RUN

Estimand: the per-fold difference in cost-adjusted return against equal weighting, abstained days held flat, across 16 rolling walk-forward folds. Margin: the median difference must exceed the interquartile range of the equal-weight arm’s own per-fold returns. Tests: a HAC-corrected t test and a block bootstrap, both at 0.05. Only the weighting function differs between arms, because every arm reads the same byte-identical cache.

The margin is scaled to dispersion rather than set as a fixed percentage, so that a gain is only called a gain when it is larger than the fold-to-fold noise it has to be read against.